My Vivino: Wine Data Analysis and Recommendation System

Experiment report | 720 wines | Python, pandas, SciPy and scikit-learn

Research question

Is there a statistically significant relationship between the price of a wine and its average customer rating? I also wanted to see whether I could use the information I collected to build a simple system that recommends similar wines.

1. Introduction

For this project, I chose wine data from Vivino because it includes both prices and customer ratings. I wanted to find out whether more expensive wines actually tend to have higher ratings, instead of just assuming they do. After exploring the data, I decided to use the same dataset for a second task: recommending wines based on their characteristics.

2. Data collection and preparation

I collected the data using a Python script and saved it as a CSV file. The final dataset has 720 wines and 13 columns, including wine name, winery, wine type, country, region, price, average rating and number of ratings. Prices are in USD. The dataset contains 120 wines in each of six types: Red, White, Sparkling, Rosé, Dessert and Fortified.

I loaded the CSV into pandas and checked column types, missing values, duplicates and numerical ranges. I found seven missing vintage-year values and kept those rows because the missing years did not affect the price-rating analysis or the recommendation features. I did not find invalid prices, ratings outside the 1–5 range or non-positive rating counts. The data was ready for exploratory analysis without dropping any wines.

3. Exploratory data analysis

I started with price and average rating. The average price was $54.28, but the median was only $32.28. A few expensive wines pushed the mean up, with the highest price reaching $487.70. The average customer rating was 3.98 out of 5, and most wines were rated between about 3.8 and 4.2.

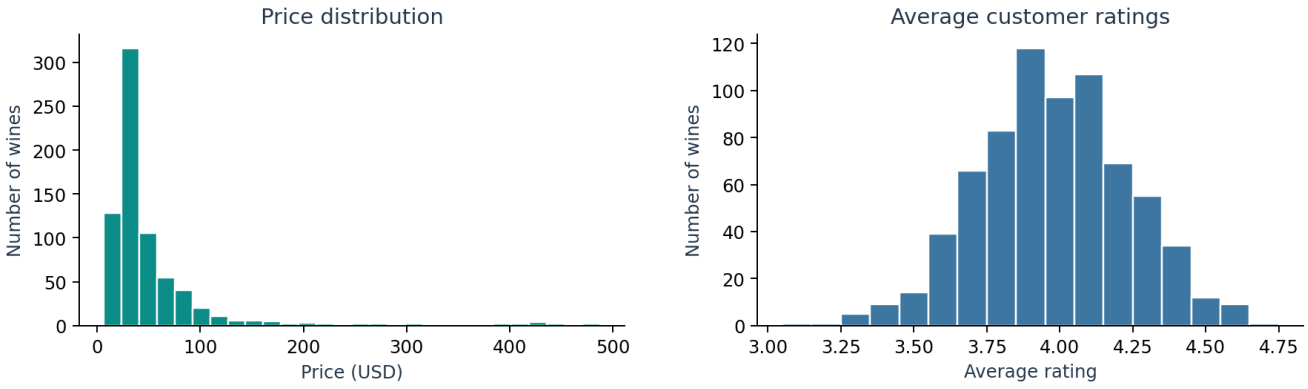


Figure 1. Price is strongly right-skewed, while customer ratings are concentrated around 4.0.

4. Price and customer ratings

The scatter plots suggested a positive relationship: higher-priced wines often had higher average ratings. However, the points were spread out, so price clearly did not explain everything. I also repeated the plot after excluding wines above the 95th price percentile (about $145.97) to check whether very expensive wines were driving the pattern.

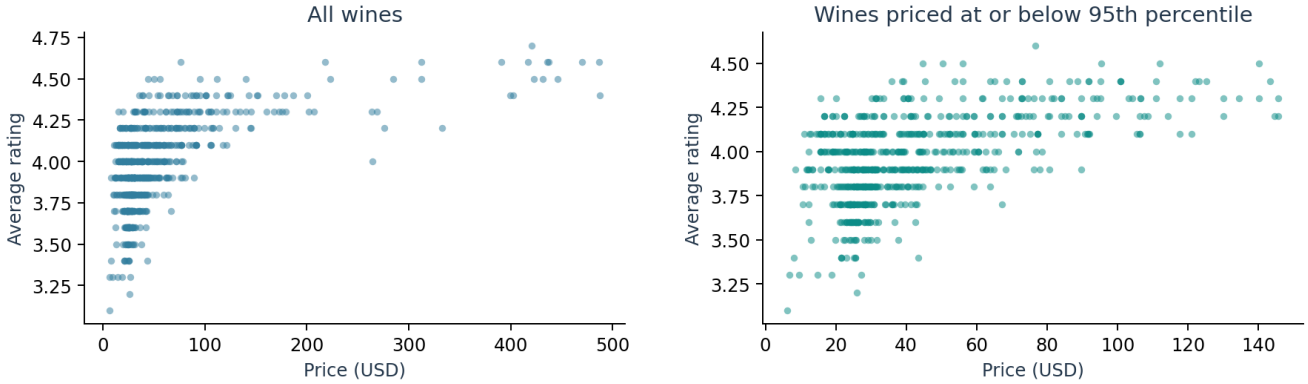


Figure 2. The positive pattern remains visible even when the top 5% of prices are excluded.

5. Hypothesis testing

Null hypothesis (H0): There is no linear correlation between wine price and average rating in the population represented by this sampling process (Pearson correlation = 0). Alternative hypothesis (H1): There is a non-zero linear correlation.

I used Pearson correlation to measure the linear relationship and Spearman correlation to check whether the overall ranked relationship was similar. I also repeated both tests after excluding the most expensive 5% of wines.

Test	Full dataset	Price <= 95th percentile
Pearson correlation	0.5503	0.5565
Spearman correlation	0.5766	0.5132

The full-data Pearson p-value was below 0.0001, so I rejected the null hypothesis of zero linear correlation. The Spearman result was also statistically significant (p about 4.99 × 10^-65). Both methods showed a moderate positive association. Since the correlations stayed positive after removing the highest-priced wines, I do not think the relationship is explained only by a few expensive bottles.

6. Wine type and country comparisons

I compared average prices and ratings across wine types. Fortified wines had the highest average price ($83.83), while Rosé had the lowest ($28.31). Dessert wines had the highest average rating (about 4.14), and Rosé had the lowest (about 3.81). These are descriptive comparisons, not evidence that wine type causes a rating to be higher or lower.

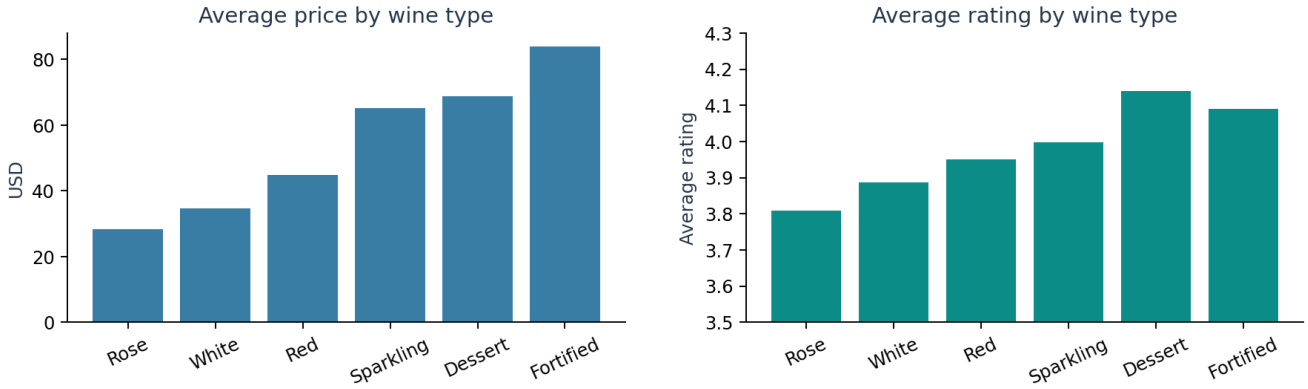


Figure 3. Mean prices vary considerably by type, while mean ratings differ by a smaller amount.

I also compared countries, but only included countries with at least ten wines. Portugal and France had higher average prices than Italy and Spain in this sample. Portugal had the highest average rating among these countries, although the rating differences were fairly small.

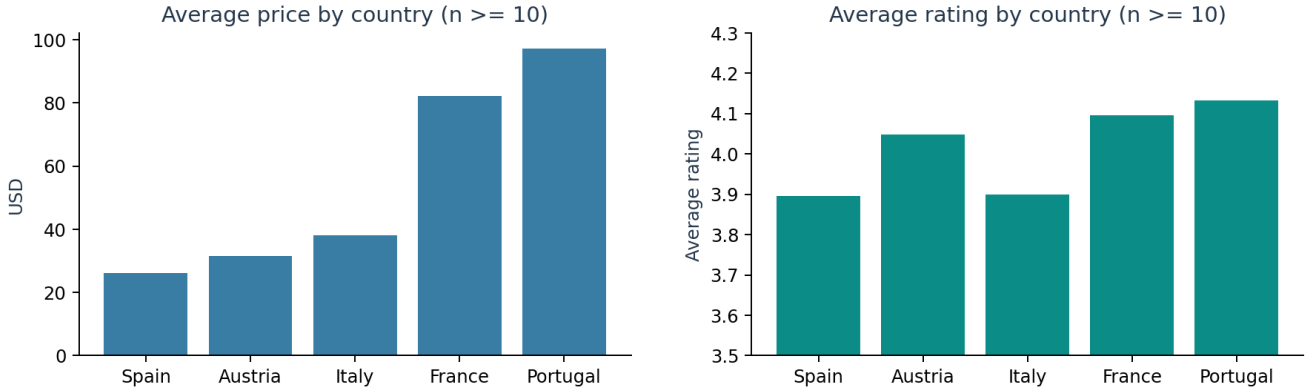


Figure 4. Country comparisons are limited to countries with at least 10 wines.

7. Do more ratings mean a higher score?

I looked at the number of customer ratings because I wanted to know whether more popular wines were also rated higher. Rating counts were very uneven: the median was about 2,743, while the maximum was 149,763. The Pearson correlation between rating count and average rating was only 0.1255, and Spearman correlation was -0.0790. After excluding the highest 5% of rating counts, the Pearson correlation was about -0.0392 and Spearman was -0.1581. These relationships are weak and sensitive to how the data is analyzed.

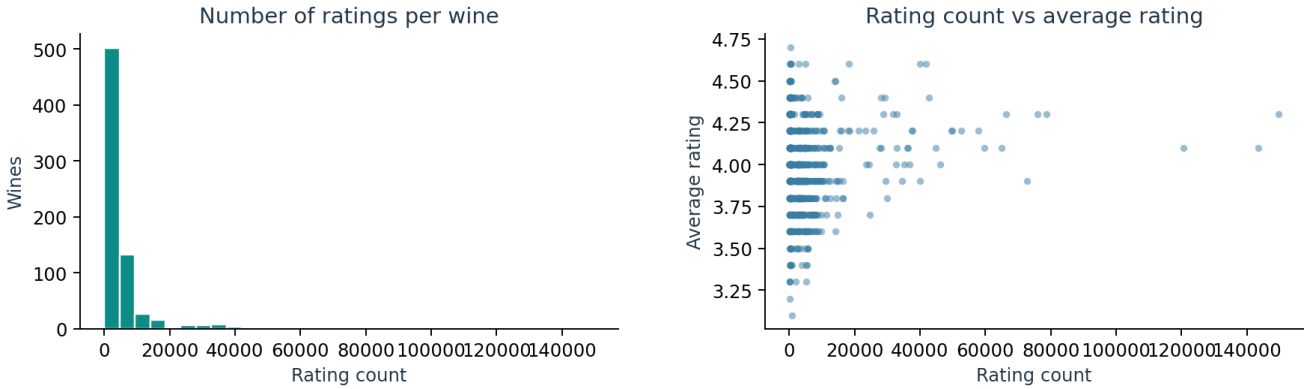


Figure 5. Most wines have relatively few ratings, and there is no strong consistent rating-count relationship.

8. Building the recommendation system

After finishing the analysis, I built a content-based recommendation system using three categorical features: wine type, country and region. I used one-hot encoding to turn these text values into numerical columns and cosine similarity to compare wines. For each selected wine, the system removes the original wine and returns the five most similar entries.

Model 1: Equal weights. I initially treated all three features equally. The model generally worked, but I noticed that it could recommend a Red wine when the selected wine was Rosé, because matching country and region could tie with matching type and country.

Model 2: Weighted features. I changed the similarity calculation to give wine type 60% importance, country 25%, and region 15%. I scaled each one-hot feature group by the square root of its weight before calculating cosine similarity. This made matching wine type more important than matching country and region together.

9. Model evaluation

I evaluated both models by requesting five recommendations for each of the 720 wines, giving 3,600 recommendations per model. My metric was the percentage of recommendations with the same wine type as the original wine.

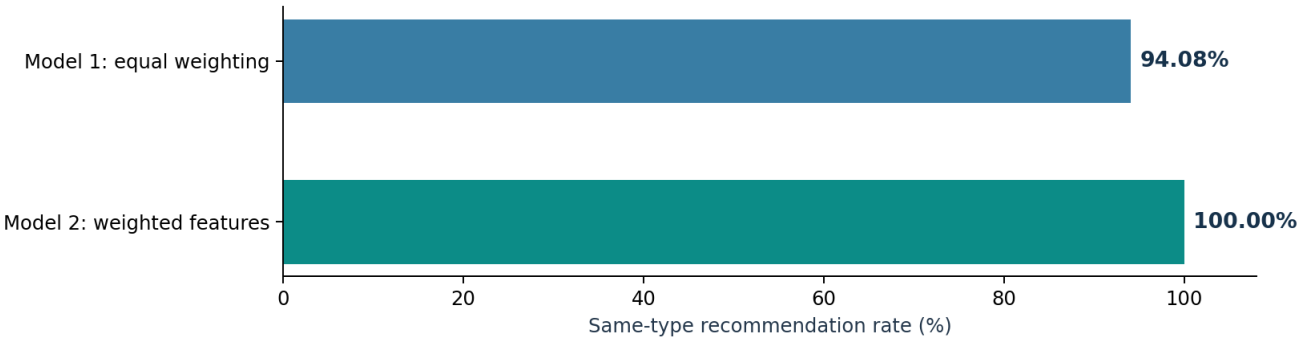


Figure 6. Model 2 improves the same-type recommendation rate by 5.92 percentage points.

Model 1 reached 94.08% and Model 2 reached 100.00%. This shows that the weighting change achieved my specific goal of prioritizing wine type. For one Rosé example, Model 1 returned three Rosé wines and two Red wines, while Model 2 returned five Rosé wines. I would not call the 100% result perfect recommendation accuracy, because I did not test whether real users would enjoy the suggestions.

10. Limitations

The biggest limitation is how the data was collected. I used 120 wines from each wine type, which makes the sample balanced by design rather than representative of all wines on Vivino. Italy also accounts for 385 of the 720 entries, so country comparisons are uneven. The dataset is relatively small, and customer ratings are subjective. The results show association, not cause and effect; factors such as reputation, type and region could influence both price and ratings.

The recommendation system has limitations too. It only compares wine type, country and region, not grape variety, flavor, sweetness, budget or personal taste. My evaluation rewards same-type matches because that is what I designed Model 2 to do. It does not measure recommendation diversity, relevance to a user or actual satisfaction. The evaluation also uses the same collected dataset rather than an independent user-feedback test.

11. Conclusion

My main finding was that wine price and average customer rating had a statistically significant, moderate positive relationship in this dataset. More expensive wines tended to be rated higher, but there were plenty of exceptions and I cannot say price caused the difference. I also learned that the number of ratings was not a strong predictor of the average rating.

For the recommendation part, I started with a basic cosine similarity model, tested it, noticed a problem and improved it by adjusting the feature weights. The second model was better at keeping recommendations within the same wine type. Overall, this project helped me practice the full data science process: collecting data, cleaning it, exploring patterns, testing a hypothesis, building a model and evaluating what the results actually mean.

Tools and data source

Python, pandas, NumPy, matplotlib, SciPy and scikit-learn. Dataset collected from Vivino and analyzed in a Jupyter Notebook. Figures in this report were generated from the project CSV. Model evaluation rates are the results recorded in the notebook.